Birzeit University
Faculty of Engineering and Technology
Electrical and Computer Engineering Department
ENCS413 – Computer Networks Laboratory

Network Review

■ Objectives

(a) Review network layering (OSI model).

(b) Introduce the different types of network devices.

(c) Review the network subnetting.

1 Introduction

1.1 Open Systems Interconnection Model (OSI Model)

The Open Systems Interconnection (OSI) model is a conceptual framework that standardizes the communication functions of a telecommunication or computing system, regardless of its underlying internal structure and technology. It describes how data moves from one device to another across a network in a structured, layered way by partitioning the communication system into seven abstraction layers.

Table 1 summarizes all seven layers from top to bottom, along with their functions, common protocols, associated devices, and the name given to the data unit (PDU) at each layer.

Why Layering?

- **Standardization:** Guarantees compatibility across software and hardware from different vendors.
- **Troubleshooting:** Divides communication into layers, making it easier to isolate and diagnose problems.
- **Modularity:** Allows each layer to evolve independently without affecting adjacent layers.
- **Design simplicity:** Decomposing the problem into well-defined layers simplifies system design.
- **Learning:** Understanding network communication as a set of discrete, well-defined layers is easier to reason about.

Table 1: OSI Model — Top-Down View (Layer 7 to Layer 1).

Layer	**Functions**	**Common Protocols / Standards**	**Typical Devices**	**Protocol Data Unit**
Application (Layer 7)	Interfaces directly with end-user applications; provides network services to software	HTTP/HTTPS, FTP, SMTP, POP3, IMAP, DNS, SNMP	Application servers, web servers, email servers	Message
Presentation (Layer 6)	Data translation, format conversion, compression/decompression, and encryption/decryption	SSL/TLS, JPEG, MPEG, ASCII, EBCDIC	Gateways, encryption devices	Message
Session (Layer 5)	Establishes, manages, and terminates communication sessions between applications	NetBIOS, RPC, PPTP, SIP	Session gateways, application servers	Message
Transport (Layer 4)	End-to-end reliable data delivery, error checking, flow control, and multiplexing	TCP, UDP, SCTP	Firewalls, load balancers	Segment
Network (Layer 3)	Logical addressing, routing, and determining the best path for data across networks	IP (IPv4/IPv6), ICMP, IPsec, OSPF, BGP, RIP	Routers, Layer-3 switches	Datagram
Data Link (Layer 2)	Node-to-node data transfer, error detection, MAC addressing, and frame delimiting	Ethernet (IEEE 802.3), PPP, HDLC, VLAN (802.1Q), ARP	Switches, bridges, NICs	Frame
Physical (Layer 1)	Transmission of raw bits over a physical medium; defines cables, connectors, and signaling	Ethernet cabling, DSL, RS-232, fiber optics	Hubs, repeaters, cables, connectors	Bit

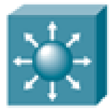
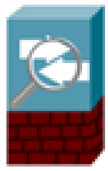
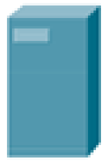
1.2 Network Devices

There are many types of network devices used in building a network topology. Table 2 summarizes the most common ones, their OSI layer of operation, and how they appear in **Cisco Packet Tracer** — the simulation tool used throughout this lab.

Table 2: Different Types of Network Devices (Part 1 of 2).

Device	**OSI Layer**	**Function**	**Packet Tracer Symbol**
Hub	Layer 1 (Physical)	Dummy device: broadcasts incoming electrical signals to all ports without any filtering or data processing. All connected devices share a single collision domain.	
Repeater	Layer 1 (Physical)	Regenerates and amplifies degraded signals to extend cable distance. Does not inspect or filter data; it simply restores signal strength.	
DSL Splitter	Layer 1 (Physical)	An analog low-pass filter that splits an incoming telephone line into two separate outputs: filtered voice (phone) and ADSL (data).	
Bridge	Layer 2 (Data Link)	Connects and filters traffic between two network segments using MAC addresses; reduces the size of collision domains.	
Switch	Layer 2 (Data Link)	Self-learning device: forwards frames based on a MAC address table; creates separate collision domains per port, significantly reducing network congestion.	
Wireless Access Point (WAP)	Layer 2 (Data Link)	Connects wireless clients to a wired network using IEEE 802.11 (Wi-Fi) standards; manages wireless medium access.	
Router	Layer 3 (Network)	Routes packets between different networks using IP addresses; uses routing protocols to determine the optimal path for each packet.	
Sniffer (Packet Analyzer)	Layers 2 & 3 (Data Link & Network)	Captures and analyzes network traffic for troubleshooting, performance monitoring, and security analysis. Inspects headers and payloads without altering traffic.	

Table 2 (cont.) — Different Types of Network Devices (Part 2 of 2).

Device	**OSI Layer**	**Function**	**Packet Tracer Symbol**
Multilayer Switch (Layer-3 Switch)	Layers 2 & 3 (Data Link & Network)	Combines switching and routing in one device: forwards frames via MAC addresses within a network, and routes packets via IP addresses between networks.	
Firewall	Layers 3 & 4 (Network & Transport)	Filters traffic based on IP addresses, port numbers, and protocols; enforces access control policies to protect network boundaries.	
Server (Web, DNS, Email)	Layer 7 (Application)	Provides application-layer services to clients: web page delivery (HTTP/HTTPS), domain name resolution (DNS), and electronic mail (SMTP/IMAP).	

Note: The icons in the "Packet Tracer Symbol" column represent the default device symbols used in Cisco Packet Tracer. You will encounter these icons throughout the simulations in this lab.

1.3 IP Subnetting

1.3.1 Key Terminology

Before discussing subnetting, the following terms must be clearly understood:

- **IP Address:** A logical address assigned to each Network Interface Card (NIC) that uniquely identifies a device on a network. There are two versions in use today:
  - **IPv4:** A 32-bit address written in dotted-decimal notation as four octets (e.g., `192.168.1.10`). It is the most widely deployed version, supporting approximately 4.3 billion unique addresses.
  - **IPv6:** A 128-bit address written in colon-separated hexadecimal groups (e.g., `2001:0db8:85a3::8a2e:0370:7334`). It was introduced to overcome IPv4 address exhaustion, providing a vastly larger address space.

  IP addresses can be assigned in two ways: **statically** by a network administrator (manually configured on the device), or **dynamically** by a **DHCP** (Dynamic Host Configuration Protocol) server, which automatically allocates an available address from a defined pool. In either case, no two interfaces on the same network may share the same IP address.
- **MAC Address (Hardware/Physical Address):** A 48-bit identifier permanently burned into the NIC by the manufacturer (also called the *burned-in address*).

It uniquely identifies network hardware at Layer 2 (Data Link) and indicates both the vendor and a unique serial number. Unlike IP addresses, MAC addresses are fixed by the hardware.

- **Network ID (Network Address):** The IP address that identifies an entire network or subnet. All bits in the *host* portion are set to **0**. This address cannot be assigned to any host.
- **Broadcast Address:** The last address in a subnet; all bits in the host portion are set to **1**. Packets sent to this address are delivered to every host in the subnet. It cannot be assigned to a host.
- **Subnet Mask:** A 32-bit value that divides an IP address into its *network* and *host* portions. Bits set to **1** identify the network part; bits set to **0** identify the host part. Written in dotted-decimal (e.g., `255.255.255.0`) or CIDR prefix notation (e.g., `/24`).
- **Wildcard Mask:** The bitwise complement of the subnet mask. Each bit that is **1** in the subnet mask becomes **0** in the wildcard mask, and vice versa. The wildcard mask indicates which bits are *variable* (the host portion). It is widely used in access control lists (ACLs) and routing protocols such as OSPF. For example, subnet mask `255.255.255.192` has wildcard mask `0.0.0.63`.
- **CIDR Prefix Length (/n):** The number of consecutive **1**-bits in the subnet mask, expressed as a slash notation (e.g., `/26` indicates 26 network bits and 6 host bits).
- **Usable Host Range:** All addresses between the Network ID and the Broadcast Address (exclusive). For a subnet with n host bits, the number of usable hosts is $2^n - 2$.
- **VLSM (Variable Length Subnet Masking):** A technique that allows different subnets within the same address space to use different prefix lengths. This enables efficient IP address allocation by sizing each subnet to match its actual host count.

1.3.2 IP Address Classes

IP addresses are organized into classes that define the default boundary between network and host bits. Only Classes A, B, and C are used for general unicast host addressing:

Table 3: Classes of Networks.

Address Class	**IP Range**	**Network Bits**	**Default Subnet Mask**
Class A	0.0.0.0 – 127.255.255.255	Leftmost 8 bits	255.0.0.0
Class B	128.0.0.0 – 191.255.255.255	Leftmost 16 bits	255.255.0.0
Class C	192.0.0.0 – 223.255.255.255	Leftmost 24 bits	255.255.255.0
Class D	224.0.0.0 – 239.255.255.255	Reserved for Multicast	
Class E	240.0.0.0 – 255.255.255.255	Reserved for Experimental Use	

1.4 How Many Networks Are There in the Figures Below?

1.4.1 Network-1 Topology

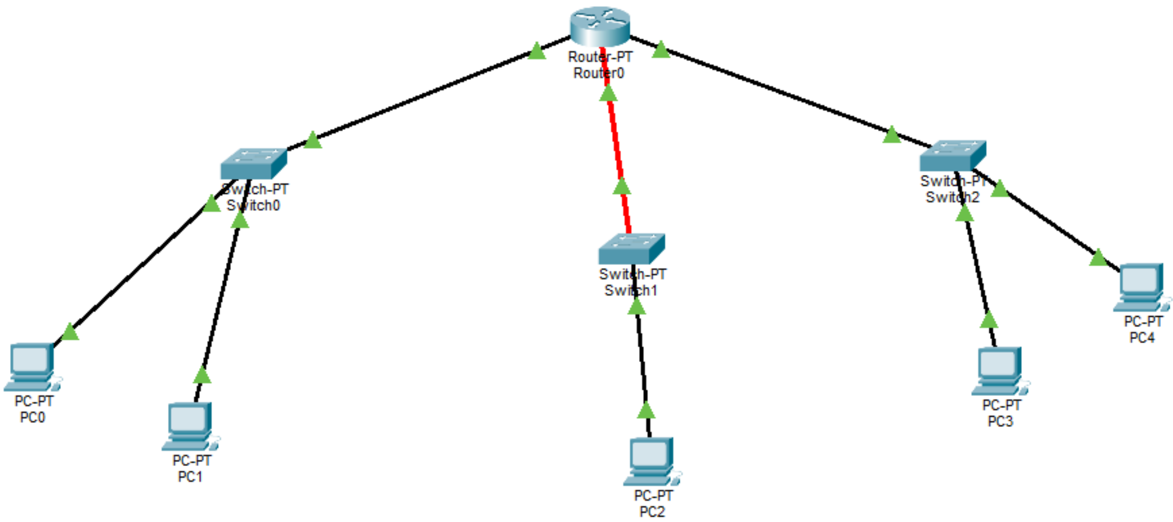


Figure 1: Network-1 Topology.

As shown in Figure 1, the router has three interfaces, and each interface connects to a separate network segment. Since each router interface defines a unique broadcast domain (and hence a unique network), the total number of networks in this figure is: **3 Networks**.

1.4.2 Network-2 Topology

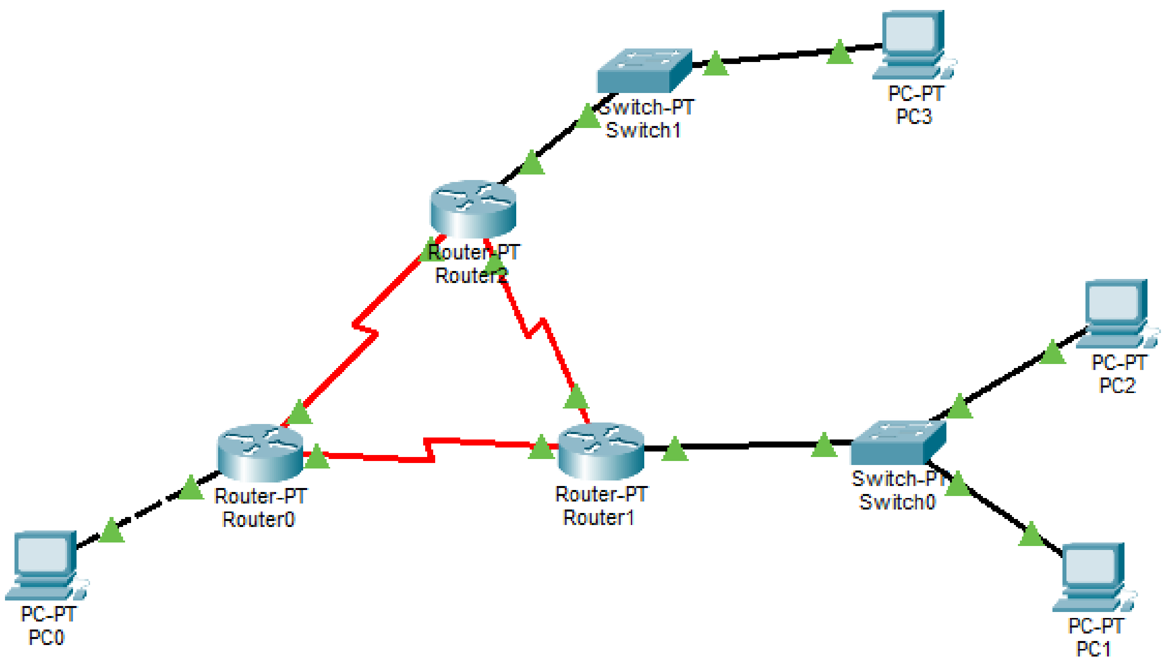


Figure 2: Network-2 Topology.

As shown in Figure 2, three routers are interconnected. Each point-to-point link between two routers constitutes a separate network (**3 inter-router networks**). Additionally, each router is connected to its own Local Area Network (**3 LAN networks**).

Thus, the total number of networks in this figure is: **6 Networks**.

1.5 Subnetting Example

The following example applies **VLSM (Variable Length Subnet Masking)** to divide the address block `192.168.1.0/24` among five networks (A, B, C, D, E) using the minimum number of IP addresses for each network.

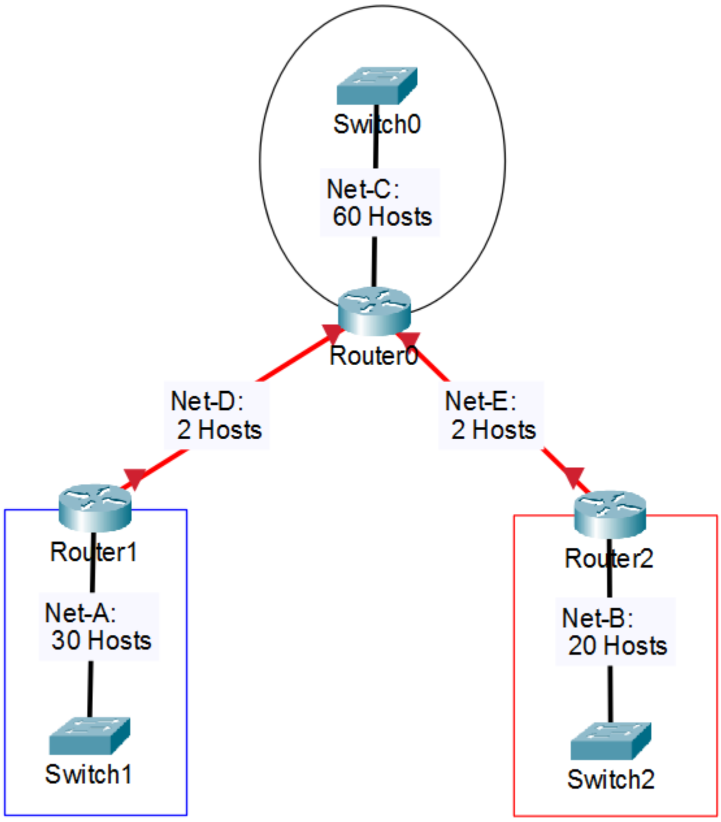


Figure 3: Example Network Topology.

From the topology, the five networks and their host requirements are:

Network	Description	Required Hosts
Net-A	Router1 ↔ Switch1 (LAN)	30
Net-B	Router2 ↔ Switch2 (LAN)	20
Net-C	Router0 ↔ Switch0 (LAN)	60
Net-D	Router0 ↔ Router1 (point-to-point)	2
Net-E	Router0 ↔ Router2 (point-to-point)	2

Step 1 — Sort Networks from Largest to Smallest

VLSM requires allocating the largest subnet first to avoid address fragmentation. Sorted order:

$$\text{Net-C (60)} \quad > \quad \text{Net-A (30)} \quad > \quad \text{Net-B (20)} \quad > \quad \text{Net-D (2)} \quad = \quad \text{Net-E (2)}$$

Step 2 — Determine the Number of Host Bits for Each Network

For each network, find the minimum n (number of host bits) such that:

$$2^n - 2 \ \geq \ \text{required hosts}$$

The subtracted 2 reserves one address for the **Network ID** (all host bits = 0) and one for the **Broadcast Address** (all host bits = 1).

Net-C — 60 hosts:

$$2^6 - 2 = 62 \geq 60 \quad \Rightarrow \quad \boxed{n = 6 \text{ host bits}} \quad \Rightarrow \quad \text{prefix /26} \;\; (32 - 6 = 26 \text{ network bits})$$

Net-A — 30 hosts:

$$2^5 - 2 = 30 \geq 30 \quad \Rightarrow \quad \boxed{n = 5 \text{ host bits}} \quad \Rightarrow \quad \text{prefix /27}$$

Net-B — 20 hosts:

$$2^5 - 2 = 30 \geq 20 \quad \Rightarrow \quad \boxed{n = 5 \text{ host bits}} \quad \Rightarrow \quad \text{prefix /27}$$

Net-D and Net-E — 2 hosts each:

$$2^2 - 2 = 2 \geq 2 \quad \Rightarrow \quad \boxed{n = 2 \text{ host bits}} \quad \Rightarrow \quad \text{prefix /30}$$

Step 3 — Assign Address Blocks (Starting from `192.168.1.0`)

Blocks are assigned sequentially. Each new subnet begins immediately after the broadcast address of the previous one.

Net-C (/26 — block size = $2^6 = 64$ addresses):

- **Network ID:** `192.168.1.0`
- **Subnet Mask:** `255.255.255.192` (last octet: 11000000_2)
- **Wildcard Mask:** `0.0.0.63`
- **Broadcast:** `192.168.1.63`
- **Usable Range:** `192.168.1.1` – `192.168.1.62` (62 hosts)
- **Next subnet starts at:** `192.168.1.64`

Net-A (/27 — block size = $2^5 = 32$ addresses):

- **Network ID:** `192.168.1.64`

- **Subnet Mask:** `255.255.255.224` (last octet: 11100000_2)
- **Wildcard Mask:** `0.0.0.31`
- **Broadcast:** `192.168.1.95`
- **Usable Range:** `192.168.1.65` – `192.168.1.94` (30 hosts)
- **Next subnet starts at:** `192.168.1.96`

Net-B (/27 — block size $= 2^5 = 32$ **addresses):**

- **Network ID:** `192.168.1.96`
- **Subnet Mask:** `255.255.255.224` (last octet: 11100000_2)
- **Wildcard Mask:** `0.0.0.31`
- **Broadcast:** `192.168.1.127`
- **Usable Range:** `192.168.1.97` – `192.168.1.126` (30 usable, 20 required)
- **Next subnet starts at:** `192.168.1.128`

Net-D (/30 — block size $= 2^2 = 4$ **addresses):**

- **Network ID:** `192.168.1.128`
- **Subnet Mask:** `255.255.255.252` (last octet: 11111100_2)
- **Wildcard Mask:** `0.0.0.3`
- **Broadcast:** `192.168.1.131`
- **Usable Range:** `192.168.1.129` – `192.168.1.130` (2 hosts)
- **Next subnet starts at:** `192.168.1.132`

Net-E (/30 — block size $= 2^2 = 4$ **addresses):**

- **Network ID:** `192.168.1.132`
- **Subnet Mask:** `255.255.255.252` (last octet: 11111100_2)
- **Wildcard Mask:** `0.0.0.3`
- **Broadcast:** `192.168.1.135`
- **Usable Range:** `192.168.1.133` – `192.168.1.134` (2 hosts)
- **Next subnet starts at:** `192.168.1.136`

Step 4 — Summary Table

Table 4: VLSM Subnetting Summary for `192.168.1.0/24`

Net	Network ID	Subnet Mask	Usable IP Range	Broadcast	Wildcard Mask
Net-C (60 Hosts)	192.168.1.0/26	255.255.255.192	192.168.1.1 – 192.168.1.62	192.168.1.63	0.0.0.63
Net-A (30 Hosts)	192.168.1.64/27	255.255.255.224	192.168.1.65 – 192.168.1.94	192.168.1.95	0.0.0.31
Net-B (20 Hosts)	192.168.1.96/27	255.255.255.224	192.168.1.97 – 192.168.1.126	192.168.1.127	0.0.0.31
Net-D (2 Hosts)	192.168.1.128/30	255.255.255.252	192.168.1.129 – 192.168.1.130	192.168.1.131	0.0.0.3
Net-E (2 Hosts)	192.168.1.132/30	255.255.255.252	192.168.1.133 – 192.168.1.134	192.168.1.135	0.0.0.3